

EFFECTS OF MONOUNSATURATED OILS ON SERUM LIPID PROFILE WHICH ONE IS BETTER?

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ABSTRACT

To compare the effects of two dietary monounsaturated oils on plasma lipid profile in Albino rats, thirty-six Albino rats were divided into three groups. Each group was fed a diet containing a different dietary oil for 12 weeks. Lipid profiles of control, rapeseed oil group and olive oil group at 0-week and 12 weeks were estimated. Results showed that in olive oil group, there was a non-significant ($P>0.05$) decrease of 3.8% in TC, significant ($P<0.005$) decrease of 12% in LDL-c, 18% in VLDL-c, 17% in TAG, 8.5% decrease in its levels. There was a significant increase of 21% in HDL-c levels. In Rapeseed oil group, there was a non-significant decrease of 26% in VLDL-c, 3.6% in TAG, 1.8% in TL levels; there was a significant increase of 11.6% in TC, 20.8% in LDL-c, 14.3% in HDL-c levels. We conclude that not all monounsaturated oils have the same effects on serum lipid profile. In this study, olive oil diet corrected the lipid profile more beneficially as compared with the rapeseed oil.

INTRODUCTION

Despite steady progress in treatment of cardiovascular diseases people are still dying of these diseases, although at later ages (Fuster 1999). By the year 2020 coronary heart disease (CHD) and stroke will hold first and fourth places, respectively, in the World Health Organization list of leading causes of disability (Napoli, 1999). A worldwide epidemic of cardiovascular diseases is evolving, and atherosclerosis, often with thrombosis superimposed, is by far the most frequent underlying cause (Murry and Lopez, 1997).

Lowering plasma cholesterol by diet and drugs slows and may even reverse the progression of atherosclerotic vascular lesions and the complications they cause (Ganong, 2003; Reaven and Witztum, 1996). To treat elevated plasma cholesterol, dietary measures are always initiated first and these only may obviate the need for drugs. In fact, most patients with elevated low-density lipoprotein

cholesterol (LDL-c) can be managed appropriately with diet in which monounsaturated fats should predominate (Malloy and Kane 1998). Recent studies suggest that monounsaturated fats may be the most healthful (Silverthorn, 2000). Comparative effects of various dietary monounsaturated oils on plasma lipids are inadequately studied. In this study, we compared the effect of two monounsaturated oils on plasma lipids in Albino rats.

MATERIALS AND METHODS

This study was conducted in the Department of Pathology, Postgraduate Medical Institute (PMGI) Lahore.

Oils:

Rapeseed oil and olive oil were studied. Rapeseed oil (Urdu: *Sarson Ka Tael*) is expressed from the seeds of *Brassica campestris* species of plants (Nandkarni, 1986). Olive oil (Urdu: *Zaitoon Ka Tael*) is

produced by extraction of ripe fruit of Olive tree (British Pharmacopeia Commission 1993). Both contain monounsaturated fatty acids (MUFA) in predominant amount. Fatty acid composition of both oils was determined by gas liquid chromatography (GLC) and it was shown that Rapeseed oil Contains 51% of a MUFA- the Erucic acid, which is a 22-carbon, omega-9 type of fatty acid having a double bond at n-13 carbon atom (ω 9, C₂₂: 1; 13). It also contains 5% saturated fatty acids [Palmitic acid (C₁₆) and Stearic acid (C₁₈)] and small amounts of Oleic acid (ω 9, C₁₃: 1;9), Linoleic acid (ω 6, C₁₃: 2; 9, 12) and alpha Linolenic acid (ω 3, C₁₈: 3; 9, 12, 15). In Olive oil, the most abundant MUFA present is oleic acid; it also contains 14% saturated fatty acids (Palmitic acid and Stearic acid) and small amounts of Palmitoleic acid (ω 7, C₁₆: 1; 9) and Linoleic acid (ω 6, C₁₃: 2; 9, 12).

Rat Diets:

Three types of rat diets were prepared which differ only in the type of oil fat. Control diet was having 5% corn oil which was a PUFA oil (having Polyunsaturated fatty acids in abundance). Rapeseed oil diet contained 20% of rapeseed oil fat and olive oil diet had 20% of olive oil fat.

Methods:

Thirty-six Albino rats were divided into groups of twelve rats each. Each group had same number of male and female rats of eight weeks age. All three groups were fed control diet for one week. After one week each group was fed only one type of diet for twelve weeks. Groups were named according to the type of diet fed i.e., Control group, Rapeseed oil group and Olive oil group. Blood samples were collected at the start of the experimental period (0-week sample) and after the completion of twelve weeks period (12-weeks sample). These samples were allowed to clot and serum was separated by centrifugation. Serum was analyzed for estimation of lipid profile. Lipid profiles in various groups were compared. Results were statistically analyzed by using student's t test.

RESULTS

Results of the study are shown in table-I. As there was non-significant difference among the lipid profiles of all the three groups at 0-week and also a non-significant difference in the 0-week and 12-weeks samples of the control group, we present here the 12-week's values of study groups with control group. In

Table
Comparison of lipid profile of study groups with control group at 12-weeks

Groups ↓	Lipid Profile					
	TC	LDL-c	HDL-c	VLDL-c	TAG	TL
Control	82.4±7.0	30.7 ± 6.4	30.1±2.1	129.3±9.4	101.7±9.1	478.8± 3.4
Olive oil group	79.2± 8.0	27.0±6.8	36.5±3.6	106.1±9.4	84.2±6.7	437.9±10.5
% Difference	3.8 ↓	12.0 ↓*	21.0 ↑*	18.0 ↓*	17.0 ↓*	8.5 ↓*
Rapeseed oil group	92.0±7.7	37.1±5.3	34.4±3.9	132.7± 10.2	105.5±6.9	487.5±12.3
% Difference	11.6 ↑*	20.8 ↑*	14.3 ↑*	2.6 ↑	3.6 ↑	1.8 ↑

↑ = increase; ↓ = decrease. Values expressed as mean ± SD in mg per 100 ml

* = P < 0.05 (significant) as compared with control group

TC = total cholesterol; LDL-C = low density lipoprotein cholesterol; HDL-C = high density lipoprotein cholesterol; VLDL-c = very low density lipoprotein cholesterol; TAG = triacylglycerol; TL = total lipids.

olive oil group, there was significant ($P<0.05$) decrease in low density lipoprotein cholesterol (LDL-c), very low density lipoprotein cholesterol (VLDL-c), triacylglycerols (TAG) and total lipids (TL); non-significant decrease in total cholesterol (TC) and significant increase ($P<0.05$) in high density lipoprotein cholesterol (HDL-c) at 12-weeks. In rapeseed oil group, there was overall increase in all lipid fractions (TC, LDL-c, HDL-c, VLDL-c, TAG and TL). The increase was significant in total cholesterol, LDL-c and HDL-c fractions at 12-weeks.

DISCUSSION

By far the most important preventive measure against the development of atherosclerosis is to eat a low-fat diet that contains mainly unsaturated fat with low cholesterol content (Guyton and Hall 2000). In our study, olive oil diet increased the HDL-cholesterol significantly and decreased the non-HDL-cholesterol (LDL-c, VLDL-c, TAG) significantly ($P<0.05$). Some workers (Mensink and Katan, 1987; Jacottot *et al.*, 1988) reported the HDL-c raising property of olive oil diet while other workers (Garg 1988, Grundy 1986) found that the olive oil diet decreased the non-HDL cholesterol significantly ($P<0.05$). Rapeseed oil diet was shown to increase the HDL-cholesterol and non-HDL cholesterol (LDL-c and TC) significantly ($P<0.05$) in our study. Similar findings were reported by (Carrol, 1962) about rapeseed oil diet. Rapeseed oil, because of erucic acid, was found to be cardiotoxic in man (Reynolds, 1989). Canola oil is the edibility of rapeseed oil which contains erucic acid in less than 2% amount.

It is suggested that monounsaturated oils decrease the level of LDL-cholesterol by up-regulation of LDL-c receptors on the cells, so increasing the uptake of LDL-c by the cells and decreasing its level in the blood. Increase in plasma HDL-c may be secondary to this decrease in LDL-c (Mayes, 1996). LDL-c receptor is a large complex molecule which is specialized for endocytosis of LDL-c via clathrin coated pits.

In this study olive oil diet decreased the level of LDL-c and rapeseed oil diet increased the plasma LDL-c level. Both oils are having predominance of monounsaturated fatty acids. Olive oil has oleic acid, an 18-carbon, ω -9 type of fatty acid; Rapeseed oil has erucic acid, a 22-carbon, ω -9 type of fatty acid. Possibly the chain-length of these fatty acids is affecting the expression of LDL-c receptors on the cells.

CONCLUSION

The general belief that all vegetable oils containing monounsaturated fatty acids are beneficial in dyslipidemia is not correct. Not all MUFA oils lower the plasma cholesterol level. In this study olive oil was found better in correcting the lipid profile as compared with the rapeseed oil. Olive oil diet not only lowers the LDL-c but also increases the HDL-c.

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